

Proposition 5.1

parabolic rescaling	Han--Li--Sun	Proposition 5.1	(X_0,T)
Andrews--Chow--Guenther--Langford	<i>Extrinsic Geometric Flows</i>	369	
$X_j(x,t)=\lambda_j\Big(X(x,t_j+\lambda_j^{-2}t)-X(x_j,t_j)\Big),\qquad \lambda_j=H(x_j,t_j).$			
Proposition 5.1	spacetime L^2		
mean curvature flow	shrinking sphere	$ H_\infty (0,0)=1$	smooth pointed
blow-up limit			
Han--Li--Sun	beta-symplectic gradient flow		fixed-
center	Proposition 5.1		
$q_j=\lambda_j\big(F(x_j,t_j)-X_0\big),\qquad \theta_j=\lambda_j^2(T-t_j).$			
q_j	θ_j	H_j,V_j,f_j	L^2
$q_j\rightarrow 0$	spacetime area bound		$F_j^\perp$
		Han--Li--Sun	$q_j^\perp$
			L^2
			moving-cutoff
(5.5)			

1.

$F:\Sigma\times[0,T)\rightarrow M$	beta-symplectic critical-surface gradient flow	normal coordinates
$\partial_\tau F=f,\qquad f=\frac{\cos^2\alpha\,H-\beta\sin^2\alpha\,V}{\cos^2\alpha+\beta\sin^2\alpha},$		
H	mean-curvature vector	V
	Han--Li--Sun	normal field
symplectic-angle bound	α	Kähler angle
$\cos\alpha\geq\delta>0.$		
Han--Li--Sun	(X_0,T)	
$G_\lambda(x,u)=\lambda\big(F(x,T+\lambda^{-2}u)-X_0\big),\qquad u<0.$		
Proposition 5.1	$R>0$	$-\infty<a<b<0$
		$\lambda\rightarrow\infty$

$$\begin{aligned} & \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |H_\lambda|^2 d\mu_u^\lambda du, \quad \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |V_\lambda|^2 d\mu_u^\lambda du, \\ & \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |f_\lambda|^2 d\mu_u^\lambda du, \quad \int_a^b \int_{\Sigma_u^\lambda \cap B_R(0)} |G_\lambda^\perp|^2 d\mu_u^\lambda du. \end{aligned}$$

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$$x_j \in \Sigma \quad t_j \nearrow T$$

$$p_j = F(x_j, t_j), \quad \lambda_j = |H|(x_j, t_j) \longrightarrow \infty,$$

$$F_j(x, t) = \lambda_j \left(F(x, t_j + \lambda_j^{-2} t) - p_j \right). \quad (1.1)$$

rescaled time interval

$$F_j(x_j, 0) = 0, \quad |H_j|(x_j, 0) = 1. \quad (1.2)$$

$$(1.1) \quad R > 0 \quad s_1 < s_2 < 0 \quad \text{Proposition 5.1}$$

2. scaling identities

$$\tau_j(t) = t_j + \lambda_j^{-2} t.$$

$$\text{translation} \quad \text{tangent} \quad \text{normal spaces} \quad \text{ambient dilation } y \mapsto \lambda_j y \quad \text{induced metric} \quad \lambda_j^2 \quad \Sigma$$

$$g_j = \lambda_j^2 g, \quad d\mu_t^j = \lambda_j^2 d\mu_{\tau_j(t)}, \quad d\tau = \lambda_j^{-2} dt. \quad (2.1)$$

second fundamental form inverse metric

$$H_j = \lambda_j^{-1} H. \quad (2.2)$$

$$\text{Kähler angle} \quad \text{dilation} \quad \text{scaled orthonormal tangent frame} \quad \lambda_j^{-1}$$

$$V_j = \lambda_j^{-1} V. \quad (2.3)$$

$$f_j = \frac{\cos^2 \alpha_j H_j - \beta \sin^2 \alpha_j V_j}{\cos^2 \alpha_j + \beta \sin^2 \alpha_j} = \lambda_j^{-1} f, \quad \partial_t F_j = f_j. \quad (2.4)$$

$\cos \alpha_j \geq \delta$ uniform pointwise estimate

$$|f_j|^2 \leq C(\beta, \delta) (|H_j|^2 + |V_j|^2). \quad (2.5)$$

fixed singular point (X_0, T)

$$q_j = \lambda_j(p_j - X_0), \quad \theta_j = \lambda_j^2(T - t_j). \quad (2.6)$$

fixed-center coordinates

$$G_j(x, t) := \lambda_j (F(x, \tau_j(t)) - X_0) = F_j(x, t) + q_j, \quad (2.7)$$

$$T - \tau_j(t) = \lambda_j^{-2}(\theta_j - t). \quad (2.8)$$

backward heat kernel

$$\rho_{X_0, T}(F, \tau_j(t)) d\mu_{\tau_j(t)} = \psi_j(F_j, t) d\mu_t^j, \quad (2.9)$$

$$\psi_j(z, t) = \frac{1}{4\pi(\theta_j - t)} \exp\left(-\frac{|z + q_j|^2}{4(\theta_j - t)}\right). \quad (2.10)$$

normal-position identity

$$G_j^\perp = (F_j + q_j)^\perp = F_j^\perp + q_j^\perp. \quad (2.11)$$

fixed-center monotonicity

$$(F_j + q_j)^\perp \quad F_j^\perp$$

3. Gaussian lower bound sharp criterion

$R > 0$ $s_1 < s_2 < 0$ cylinder

$$|F_j| \leq R, \quad s_1 \leq t \leq s_2$$

j $c_R > 0$ $\psi_j \geq c_R$

$$\sup_j |q_j| < \infty, \quad \sup_j \theta_j < \infty. \quad (3.1)$$

$$|q_j| \leq Q \quad 0 \leq \theta_j \leq \Theta$$

$$-s_2 \leq \theta_j - t \leq \Theta - s_1, \quad |F_j + q_j| \leq R + Q,$$

$$\psi_j(F_j, t) \geq \frac{1}{4\pi(\Theta - s_1)} \exp\left(-\frac{(R + Q)^2}{4(-s_2)}\right) > 0. \quad (3.2)$$

$\theta_j \rightarrow \infty$ $z = 0, t = s_2$ prefactor $|q_j| \rightarrow \infty$ θ_j exponential factor
 ball (3.1) uniform Gaussian lower bound

4. “ ”

4.1 shrinking sphere

$M_t = S^n_{r(t)} \subset \mathbb{R}^{n+1}$ shrinking round sphere

$$r(t) = \sqrt{2n(T-t)}.$$

$e \quad t_j \nearrow T \quad p_j = r(t_j)e$

$$\lambda_j = |H|(p_j,t_j) = \frac{n}{r(t_j)}.$$

(1.1) j rescaled flow sphere center $-ne$ radius

$$r_j(t) = \sqrt{n^2-2nt},$$

$$|H_j|(t) = \frac{n}{\sqrt{n^2-2nt}}, \qquad |H_j|(x_j,0) = 1. \tag{4.1}$$

$\varepsilon > 0 \quad s_1 = -\varepsilon \quad s_2 = -\varepsilon/2$

$$R > \sqrt{n^2+2n\varepsilon} - n,$$

$t \in [s_1,s_2]$ sphere $B_R(0)$ spherical cap

$$\int_{s_1}^{s_2} \int_{M_t^j \cap B_R(0)} |H_j|^2 \, d\mu_t^j \, dt$$

j Sun--Jun surface dimension $n=2$

4.2 smooth nonflat limit obstruction

F_∞ round symmetry moving-center rescalings smooth locally converges
normalization

$$|H_\infty|(p_\infty,0) = 1.$$

compact coordinate patch K $s_1 < s_2 < 0$

$$|H_\infty| \geq \frac{1}{2} \quad \text{on } K \times [s_1,s_2].$$

smooth convergence j $|H_j| \geq 1/4$ induced area density patch R

$F_j(K,t) \subset B_R(0)$ $c > 0$

$$\liminf_{j \rightarrow \infty} \int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 \, d\mu_t^j \, dt \geq c > 0. \tag{4.2}$$

beta-symplectic curvature-normalizing sequence

smooth nonflat blow-up limit

$$|H_j|(x_j, 0) = 1$$

compactness

$$(4.2)$$

regularity

5. beta-symplectic flow

Han--Li--Sun fixed-center Proposition 5.1

Theorem 5.1 moving-center transfer

$$\begin{array}{llllll} F & \text{beta-symplectic gradient flow} & (X_0, T) & \text{fixed singular point} & \lambda_j \rightarrow \infty & (1.1) \quad (2.6) \\ F_j, q_j, \theta_j & & & & & \end{array}$$

1. λ_j fixed-center rescalings

$$\tilde{F}_j(x, u) = \lambda_j (F(x, T + \lambda_j^{-2} u) - X_0)$$

Han--Li--Sun Proposition 5.1

fixed-center

spacetime L^2

2. $\sup_j |q_j| < \infty \quad 0 \leq \theta_j \leq \Theta < \infty$

3. $[s_1, s_2] \subset (-\infty, 0) \quad j$ moving rescaled time domain

$$R > 0$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |H_j|^2 d\mu_t^j dt \rightarrow 0, \quad (5.1)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |V_j|^2 d\mu_t^j dt \rightarrow 0, \quad (5.2)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |f_j|^2 d\mu_t^j dt \rightarrow 0. \quad (5.3)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |F_j^\perp|^2 d\mu_t^j dt \rightarrow 0, \quad (5.4)$$

$$\int_{s_1}^{s_2} \int_{\Sigma_t^j \cap B_R(0)} |q_j^\perp|^2 d\mu_t^j dt \rightarrow 0. \quad (5.5)$$

$$(5.5) \quad \text{normal-displacement condition} \quad \text{Han--Li--Sun} \quad \text{moving-cutoff} \quad (5.5)$$

$$q_j \rightarrow 0 \quad (5.6)$$

$$A_R < \infty$$

$$\int_{s_1}^{s_2} \mu_t^j(\Sigma_t^j \cap B_R(0)) dt \leq A_R, \quad (5.7)$$

$$(5.4) \quad \text{moving-center}$$

Proof

moving time t

$$u = t - \theta_j.$$

$$\theta_j = \lambda_j^2(T - t_j)$$

$$T + \lambda_j^{-2}u = t_j + \lambda_j^{-2}t.$$

$$\widetilde{F}_j(x, u) = F_j(x, t) + q_j. \quad (5.8)$$

$$\text{translation} \quad H_j, V_j, f_j \quad \text{normal spaces} \quad \text{induced measure} \quad |F_j(x, t)| < R \quad |q_j| \leq Q$$

$$|\widetilde{F}_j(x, u)| < R + Q.$$

$$0 \leq \theta_j \leq \Theta \quad t \in [s_1, s_2]$$

$$u \in [s_1 - \Theta, s_2] \subset (-\infty, 0).$$

integrands

$$\int_{s_1}^{s_2} \int_{\{|F_j| < R\}} |H_j|^2 \leq \int_{s_1 - \Theta}^{s_2} \int_{\{|\widetilde{F}_j| < R + Q\}} |\widetilde{H}_j|^2.$$

$$\text{fixed-center Proposition 5.1} \quad (5.1) \quad \text{inclusion} \quad \text{time shift} \quad (5.2) \quad (5.3)$$

$$(2.5) \quad (5.1) \quad (5.2) \quad (5.3)$$

$$\text{normal-position term} \quad (2.11)$$

$$\widetilde{F}_j^\perp = F_j^\perp + q_j^\perp.$$

$$\text{fixed-center Proposition 5.1} \quad \text{ball/time inclusion}$$

$$\int_{s_1}^{s_2} \int_{\{|F_j| < R\}} |F_j^\perp + q_j^\perp|^2 \rightarrow 0. \quad (5.9)$$

$$|F_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|q_j^\perp|^2$$

$$(5.5) \Rightarrow (5.4)$$

$$|q_j^\perp|^2 \leq 2|F_j^\perp + q_j^\perp|^2 + 2|F_j^\perp|^2$$

(5.9) (5.4) $\Rightarrow$ (5.5) orthogonal projection

$$\int |q_j^\perp|^2 \leq |q_j|^2 \int d\mu_t^j dt.$$

(5.6) (5.7)

6.

q_j θ_j shifted normal-position limit $F_j^\perp$ limit affine plane

$$F_j(u,v,t) = (u,v,-1) \subset \mathbb{R}^3, \qquad q_j = (0,0,1).$$

$$H_j = V_j = f_j = 0$$

$$F_j + q_j = (u,v,0)$$

tangent $(F_j + q_j)^\perp = 0$

$$F_j^\perp = (0,0,-1).$$

$$R > 1 \quad s_1 < s_2$$

$$\int_{s_1}^{s_2} \int_{F_j \cap B_R(0)} |F_j^\perp|^2 d\mu dt = (s_2 - s_1) \pi (R^2 - 1) > 0.$$

center displacement normal component “ ”

7. fixed cutoff

transfer proof fixed-center Proposition 5.1 ball inclusion time translation weighted
monotonicity cutoff ϕ

$$\phi(F) = \phi(X_0 + \lambda_j^{-1}(F_j + q_j)) \, .$$

$|F_j| \leq R \quad |q_j| \leq Q \quad \lambda_j^{-1}(F_j + q_j) \rightarrow 0$ uniformly j cutoff (3.2)
Gaussian weight H_j, V_j unweighted estimates (2.5) f_j monotonicity square (2.11)
shifted normal-position estimate moving cutoff

8.

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- “ Proposition 5.1 ” shrinking sphere ordinary MCF

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|----|---------------------------------|------------------------------|-----------------------|
| 3. | beta-symplectic flow | | beta-symplectic |
| | q_j, θ_j | center-control hypotheses | fixed-center proof |
| | smooth nonflat normalized limit | | moving-center version |
| 4. | q_j, θ_j | fixed-center Proposition 5.1 | $q_j^\perp$ |
| | $q_j \rightarrow 0$ | local area bound | L^2 |
| 5. | blow-up | fixed-center scaling | tangent behavior |
| | moving-center scaling | | curvature-normalizing |
| | | nonflat singularity model | |

1. B. Andrews, B. Chow, C. Guenther, and M. Langford, *Extrinsic Geometric Flows*, Graduate Studies in Mathematics 206, American Mathematical Society, 2020, especially printed pp. 369 and 372--374. DOI: 10.1090/gsm/206.
2. X. Han, J. Li, and J. Sun, *Gradient flow for beta-symplectic critical surfaces*, Ann. Inst. H. Poincaré Anal. Non Linéaire 41 (2024), 1083--1116, especially Proposition 5.1. DOI: 10.4171/AIHPC/100.